Mathematics for ML: Foundations of Probability

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Outline

About Me

Yogesh Haribhau Kulkarni

Bio:

- 20+ years in CAD/Engineering software development
- Got Bachelors, Masters and Doctoral degrees in Mechanical Engineering (specialization: Geometric Modeling Algorithms).
- Currently doing Coaching in fields such as Data Science, Artificial Intelligence Machine-Deep Learning (ML/DL) and Natural Language Processing (NLP).
- Feel free to follow me at:
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About Content

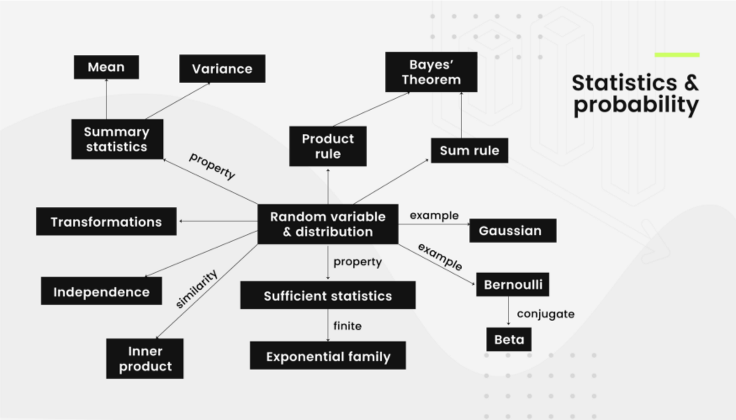
- Bare essential mathematics needed for Machine/Deep Learning
- Just to get you started.
- Each field within Machine/Deep Learning can go extremely deep.
- This is to give you enough foundation needed.

About Me

- I am not a mathematician.
- And this course won't turn YOU into one!!! (if you are not already)
- But, I will surely attempt to make you get interested in the learning, more.

Statistics

Landscape: Statistics



(Ref: The NOT definitive guide to learning math for machine learning - Favio Vazquez)

Introduction to Probability

Why Probability?

- Real life scenarios are inherently random or too complex to be completely known.
- If you knew every "Physics" aspect of motion of a dice, you could predict the outcome, deterministically.
- However this can never be done in practice, so there is a CHANCE of something happening.

Probability: the Likeliness

- How likely something is to happen.
- Many events can't be predicted with total certainty.
- The best we can say is how likely they are to happen, using the idea of probability.
- Probability is the fraction of time that outcome would occur with repeated experiments.

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Probability: Example

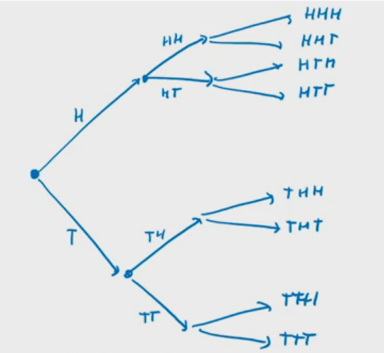
- Probability (event) = Number of ways it can happen / Total number of outcomes
- Example: the chances of rolling a "4" with a die
- Number of ways it can happen: 1 (there is only 1 face with a "4" on it)
- Total number of outcomes: 6 (there are 6 faces altogether)
- Probability of this event = 1/ 6

Probability: Example

- There are 5 marbles in a bag: 4 are blue, and 1 is red.
- What is the probability that a blue marble gets picked?
- Number of ways it can happen: 4 (there are 4 blues)
- Total number of outcomes: 5 (there are 5 marbles in total)
- Probability of this event = 4 / 5

Probability: Example

- Flip 3 coins. What is the probability that we get exactly two heads?
- Flip 1 coin, you get either H or T. Then flip 2nd coin, draw further branches (like below)
- At the end you will have 8 possible outcomes



(Ref: Math for Machine Learning - AWS Brent Werness)

- What we want are HHT, HTH, THH. So 3 out of 8.
- Probability of this event = 3 / 8

Probability

Definition

Given an experiment, let S be the set of all possible outcomes (S is often called the sample space) and A be the event of some particular outcomes (so $A \subseteq S$). The probability of A, denoted by $P(A)$, is the relative proportion of A in S.

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Probability

Example

Suppose we are rolling a fair die, then $S = \{1, 2, 3, 4, 5, 6\}$. Let A be the event that the outcome is an even number, that is, $A = \{2, 4, 6\}$. Then $P(A) = 1/2$.

Example

Suppose we are rolling two fair dice, and let A be the event that the sum of outcomes is either 6 or 7. Then $P(A) = 11/35$.

Some terminologies

- Experiment or Trial: an action where the result is uncertain. Example: Tossing a coin, throwing dice are examples of experiments.
- Outcome: A single possibility from the experiment. E.g. $\Omega = HHT, THT, \ldots$, each of these are outcomes.

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Event

- Event: a set of outcomes/results of an experiment
- An event is what we are looking for ue $E = ExactlyTwoHeads$. E is any subset of the sample space. The set of all events associated with a given experiment is called the event space.
- Example Events:
  - Getting a Tail when tossing a coin is an event
  - Rolling a "5" is an event.
- An event can include one or more possible outcomes:
  - Choosing a "King" from a deck of cards (any of the 4 Kings) is an event
  - Rolling an "even number" (2, 4 or 6) is also an event

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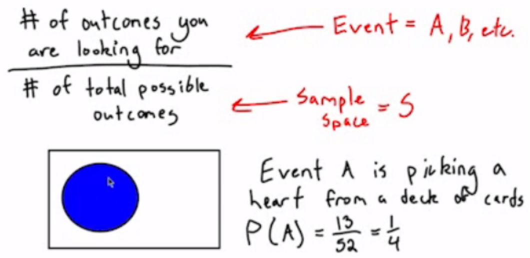
Sample space

- Sample Space: all the possible outcomes of an experiment. Example: choosing a card from a deck
  - There are 52 cards in a deck (not including Jokers)
  - So the Sample Space is all 52 possible cards: Ace of Hearts, 2 of Hearts, etc.
- A Sample Point is just one possible outcome.
- And an Event can be one or more of the possible outcomes.
- Flipping the coin: the sample space is $\{H, T\}$.
- Whats the sample space of number of heads if we flip a coin 10 times?
- $\{0, 1, \ldots, 10\}$

Probability of an event?

- Both ‘event’ and ‘probability’ are intuitively understood by most people, but we need to establish certain rules.
- ‘Rain tomorrow’, ‘3 or fewer cyclones next year’, ‘Crop yield will exceed a given threshold’ are all examples of ‘events’ whose ‘probabilities’ might be of interest.



Event probabilities

An **event** is a set of one or more points in the sample space. The probability of an event is the sum of the probabilities for the points in the event.
Example: Consider this distribution:

x	$P(X = x)$
Favorable	0.3
Neutral	0.5
Unfavorable	0.2

The response being either favorable or neutral is an event, and this event has probability $0.3 + 0.5 = 0.8$.

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Cumulative probabilities

If the sample space is ordered, then the probability of observing a value less than or equal to a given number x is the **cumulative probability at** x.

Example: If we are interested in the probability distribution of the number of times a person living in Michigan has visited Florida, we might have the following probabilities and cumulative probabilities:

x	0	1	2	3	4	5	$\cdots$
$P(X = x)$	0.3	0.1	0.2	0.1	0.06	0.08	$\cdots$
$P(X \leq x)$	0.3	0.4	0.6	0.7	0.76	0.84	$\cdots$

Independence

Two random variables are **independent** if the probability that both of them satisfy some condition at the same time is the product of the probabilities of the conditions being satisfied separately.
For example,

$$P(X = 2 \text{ and } Y = 3) = P(X = 2) \cdot P(Y = 3)$$

$$P(0 \leq X < 1 \text{ and } Y > 4) = P(0 \leq X < 1) \cdot P(Y > 4)$$

Example: Independence

- College students at an educational institution where 52% of the students are female,
- 40% of the students visit a doctor at least once a year.
- the probability that a randomly selected student is female and has visited a doctor in the last year is?
- $0.52 \times 0.4 = 0.208$.
- Corollary: if the value sis not 0.208 then these variables are **dependent**.

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Independence

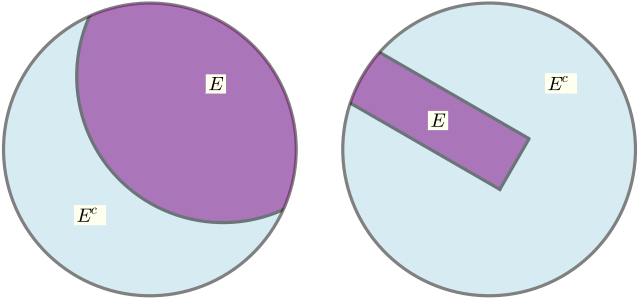
- If the proportion is greater than 0.208, the variables are **positively dependent**.
- If the proportion is less than 0.208, the variables are **negatively dependent**.

Complements

- For a particular event E, there is a related event denoted E^c which is the event that E does not happen.
- Example: E may be that the person who answers is female.
- The complementary event would be the event that the person who answers is male.
- The probability of the complement of an event is 1 minus the probability of the event: $P(E^c) = 1 - P(E)$.

Complements

Example: If the probability that a person living in Michigan has visited Florida at most two times is 0.6, the complementary event is that the person has visited Florida three or more times, and the probability of this event is 0.4.



Disjoint events

Two events are **disjoint** if they contain no points in common. These two events are disjoint:

E1:	A person's birthday is on the 31st day of the month
E2:	A person's birthday is in June

Disjoint events

But these events are not disjoint:

E1:	A person's birthday is on a Tuesday
E2:	A person's birthday is in June

Disjoint events

Events $E1$ and $E2$ are disjoint on the left, but not on the right. The large gray circle represents the full sample space.

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Disjoint events

If events $E1$ and $E2$ are disjoint, then the probability of at least one of them happening is the sum of their probabilities:

$$P(E1 \text{ or } E2) = P(E1) + P(E2).$$

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Example: Disjoint events

Roll a fair die, and consider these events:

E1:	The result is even
E2:	The result is a 3 or a 5

The event "$E1$ or $E2$" is the event that I get a 2, 3, 4, 5, or 6. This event has probability $5/6$. Since $P(E1) = 1/2$ and $P(E2) = 1/3$, we can check that $P(E1 \text{ or } E2) = 5/6 = 1/2 + 1/3 = P(E1) + P(E2)$.

Disjoint events

Now consider these events:

E1:	The result is even
E2:	The result is less than or equal to 3

These events are not disjoint, since 2 is in both events.
The event "$E1$ or $E2$" is the event that I get a 1, 2, 3, 4, or 6. This event has probability 5/6. But $P(E1) = 1/2$ and $P(E2) = 1/2$ so
$P(E1) + P(E2) \neq P(E1 \text{ or } E2)$.

Exercises

Suppose we are considering a population of students that is 52% female, and for which 40% of the students have visited a doctor in the last year.

- If these events are independent, what is the probability that a randomly sampled student is a male who has not seen a doctor in the last year?
- If these events are independent, what is the probability that a randomly sampled student is either a male who has not seen a doctor in the last year, or is a female who has not seen a doctor in the last year?
- If there are 1000 students in our population, and gender is independent of whether a student visited a doctor in the past year, how many students are females who have not seen a doctor in the past year?

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Answers

- $P(Student = Male) = 0.48$, and $P(StudentNoSeenDoctorLastYear) = 0.6$ (both are complements of the events given in the exercise). By independence, $P(Student = Male, And, StudentNoSeenDoctorLastYear) = 0.48 * 0.6 = 0.288$.
- Simalalry $P(Student = Female, And, StudentNoSeenDoctorLastYear) = 0.52 * 0.6 = 0.312$. These events are disjoint, so the probability of one or the other occuring is $0.288+0.312 = 0.6$. It's not a coincidence that this is equal to 1-0.4, the probability of visiting a doctor regardless of gender.
- The probability that a student is a female who has not seen a doctor in the last year is 0.312, so the number of such people is 1000*0.312 = 312.

Axioms* of Probability

- Probability is between 0 and 1. ie $P(E) \in [0,1]$
- Outcome is always from the sample space.
- Summation of probabilities of all possible outcomes is 1

* Axioms are statements considered to be true without any proof.

Properties of Probability

Let S be the sample space and A, B be events. Then

- $P(S) = 1, P(\emptyset) = 0$.
- If $A \subseteq B$, then $P(A) \leq P(B)$. In particular, $0 \leq P(A) \leq 1$.
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. In particular, if A and B are disjoint, then $P(A \cup B) = P(A) + P(B)$.
- $P(A^c) = 1 - P(A)$.

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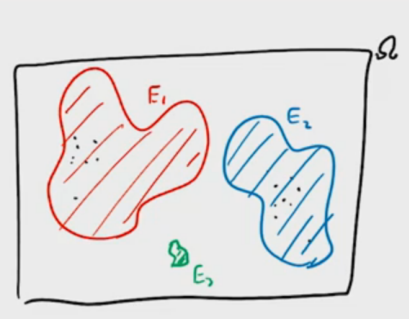
Visualizing Probability

- Let rectangular region being of entire sample space. Area is “1”.
- Points are outcomes
- Events are collection of outcomes, so sub-regions.
- Probability is shown as area of the sub-region.



(Ref: Math for Machine Learning - AWS Brent Werness)

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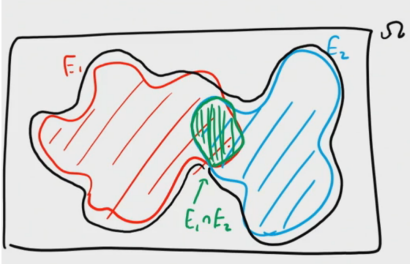
Visualizing Probability

- Union of Events is sum of probability when they are Exclusive.
- If they are not, meaning there is some intersection between events, then total probability is sum of both minus the intersection (as it got included twice)
- For 3 events summation, add all 3, minus 3 double overlaps, plus add single triple overlap.



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Probability of multiple variables

Joint Probability

- To denote the probability of multiple variables AT THE SAME TIME.
- Say, two variables, gender and hair-length. $P(male, short)$ is the probability that a person is male and has short hair.

Conditional Probability

- To denote the probability of one event AFTER THE OTHER HAS OCCURED.
- $P(longhair|male)$ is the probability of having long hair given that a person is male.

Marginal Probability

- To denote the probability of one event IRRESPECTIVE OF OTHER EVENTS.
- Probability of person being male is given by $P(male)$. It doesn't matter whether or not the person has short hair or long.

Relation between joint, conditional and marginal probabilities

- $P(A, B) = P(B|A)P(A)$
- $P(male, longhair) = P(longhair|male)P(male)$
- That is, probability of a person being male and having long hair = probability of person being male times probability of having long hair given person is male.
- Bayes theorem is a relation between conditional and marginal probabilities of two variables $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$

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Probability Formulation

Unions and intersections

- The union of two events A and B consists of everything included in A or B or both. Let
  - A = rain tomorrow
  - B = rain the day after tomorrow
  - C = 3 or fewer cyclones
  - D = 4 or 5 cyclones
- Then
  - $A \cup B$= rain in the next 2 days
  - $C \cup D$ = 5 or fewer cyclones
- $P(C \cup D) = P(C) + P(D)$, because C and D are mutually exclusive (they don't overlap).

Unions and intersections

- $P(A \cup B) \neq P(A) + P(B)$ because A and B do overlap.
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- $P(A \cap B)$ is the intersection of A and B; it includes everything that is in both A and B, and is counted twice if we add P(A) and P(B).

Single draw from a deck of 52 cards

S: all 52 cards

Event A: draw a heart

Event B: draw a face card

Event C: draw a spade

A or B $\Rightarrow$ union of A and B $\Rightarrow A \cup B$

$P(A \text{ or } C) = \frac{26}{52}$ = add probabilities of mutually exclusive events $= \overset{A}{\frac{13}{52}} + \overset{C}{\frac{13}{52}} = \frac{26}{52}$

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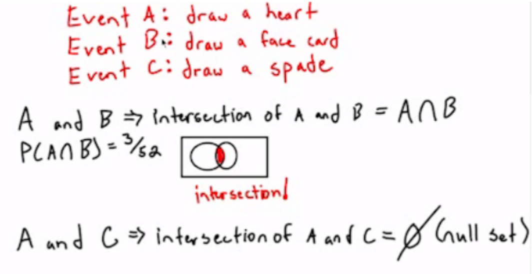
Unions and intersections

- In our example
  - $A \cap B =$ (rains tomorrow and the day after tomorrow).
  - $C \cap D$ is empty - it is impossible for C and D to occur simultaneously, so $P(C \cap D) = 0$.



Joint Probability

- Multiple events and we are trying to find chance of happening all of them together
- So? Intersection set of all events
- If A and C are independent, $P(A \cup C) = P(A)P(C)$.

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Independence

- Two events are independent, if the occurrence of one does not affect the probability of occurrence of the other.
- Similarly, two random variables are independent if the realization of one does not affect the probability distribution of the other.
- Example: rolling dice. First outcome, does not influence second outcome at all.
- In pack of cards: pick a card and then pick a card again, are independent if the first card is put back.

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Independence

- In dice example
  - $A =$ getting 3
  - $B =$ getting 6
  - $C =$ getting even number
- Total outcome of 2 events, 6x6 pairs of possible outcomes, e.g. 1-1, 1-2,1-3 . . . , 6-6.
- $P(A \cap B) = P(A)P(B) = 1/6 \times 1/6 = 1/36.$
- $P(A \cap C) = P(A)P(C) = 1/6 \times 3/6 = 3/36.$
- **NOTE: A and C are not outcome of same event, which would have been NULL. They are two separate events, one after another, thus there is non-zero probability**

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Without replacement?

- First pair pulled is black
- Second pair getting black without putting the first one back.
- These are Dependent events. So answer is not pure multiplication, but with some adjustment.
- Conditional Probability is for Dependent probabilities

Sock Drawer

8 pairs black socks
5 blue
7 white

$$P(A) = 8/20$$

$$P(A \cap B) = 8/20 \times 8/20 = 16\%$$

$$8/20 \times 7/19 = 56/380$$

Conditional probability

Single draw from a deck of 52 cards

Events

A: Draw a Heart $P(A) = \frac{13}{52}$

B: Draw a Face Card $P(B) = \frac{12}{52}$

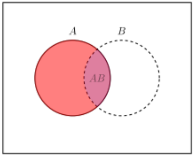
$$\frac{3}{13} = P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{3/\cancel{52}}{13/\cancel{52}}$$

$$\frac{3}{12} = P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{3/\cancel{52}}{12/\cancel{52}}$$

- Calculate probabilities of A and B, independently.
- but then if I say, that the first card is Heart, then whats the probability of Face Card.
- There are 13 hearts. Among hearts there are 3 face cards. So, 3 /13. 3 face-heart cards, total 13 hearts.
- reverse, first face, then whats prbability of getting heart. So, 12 face cards and 3 out of them are hearts. So, 3/12.

Conditional probability

- If we know that one event has occurred it may change our view of the probability of another event. Let
- A = rain today, B = rain tomorrow.
- It is likely that knowledge that A has occurred will change your view of the probability that B will occur.
- We write $P(B|A) \neq P(B)$, $P(B|A) = P(A \cap B)/P(A)$ denotes the conditional probability of B, given A.



Conditional Probability using Venn Diagram

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	A	B	C	Total
Male	8	18	13	39
Female	10	4	12	26
Total	18	22	25	65

If one student was chosen at random, find the probability that the student got an A given that he is male.

$P(\text{A}|\text{male}) = \frac{8}{39}$

Find the probability that a student is male given that he got an A.

$P(\text{male}|\text{A}) = \frac{8}{18}$

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Conditional Probability Formula

If events A and B are not independent, then $P(A \text{ and } B) = P(A) \cdot P(B|A)$.
If you pull 2 cards out of a deck, what is the probability that both are twos?

The probability that the first card is a two is $\frac{4}{52}$.

The probability that the second card is a two, given that the first was a two, is $\frac{3}{51}$.

The probability that both cards are twos is $\frac{4}{52} \cdot \frac{3}{51} = \frac{12}{2652} \simeq 0.00452$.

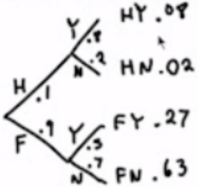
Graphical Conditional Probability

You run a record label whose A&R department is responsible for predicting whether albums by new artists will be hits or flops.

For new artists whose first album will be a hit, your A&R dept. will correctly predict the hit 80% of the time. Unfortunately, when the album is destined to be a flop, the A&R dept. will still predict a hit 30% of the time.

You know that debut albums by new artists will turn out to be hits just 10% of the time. The rest of the debut albums will be flops.

Q: If your A&R dept. predicts a debut album will be a hit, what are the chances that it really will be?



$$\frac{.08}{.08+.27} = \frac{.08}{.35} = .229 = 22.9\%$$

Bayes Theorem Example

Bayes Theorem Example

- Suppose that 0.8% of the general population has a certain disease, and suppose that we have a test for it.
- The test says so 90% of the time.
- If a person does not have the disease, the test gives a false diagnosis 7% of the time.
- When a particular patient tests positive, what is the probability they have the disease?

Bayes Theorem Example

- Let D be the event that the person has the disease and $\bar{D}$ be its complement;
- let Y (for "yes") be the event that the test *says* the person has the disease, with complement $\bar{Y}$.

$$\begin{aligned} P(D) &= 0.008 \\ P(Y|D) &= 0.90 \\ P(Y|\bar{D}) &= 0.07. \end{aligned}$$

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Bayes Theorem Example

Let's find the probability of Y by itself. Using the partition theorem we know this is

$$\begin{aligned} P(Y) &= P(Y|D)\,P(D) + P(Y|\bar{D})\,P(\bar{D}) \\ &= 0.90 \cdot 0.008 + 0.07 \cdot 0.992 \\ &= 0.077. \end{aligned}$$

So the non-conditional probabilities are

$$P(\bar{D}) = 0.992 \qquad P(\bar{Y}) = 0.923$$

$$P(D) = 0.008 \qquad P(Y) = 0.077.$$

Bayes Theorem Example

- Looking at D and Y separately,
- We can think of the population at large as being split into those with and without the disease, and those for whom the test is positive or negative.
- Suppose in particular that we have a sample of 1000 people who are representative of the general population.

Bayes Theorem Example

Here are some (very rectangular) Venn diagrams:

$\bar{D}$	D	
992	8	

923	$\bar{Y}$
77	Y

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Bayes Theorem Example

Bayes' theorem has to do with how these two partitions intersect to make four groups:

$\bar{D}$	D	
$P(\bar{Y}\|\bar{D})\cdot 992 =?$	$P(\bar{Y}\|D)\cdot 8 =?$	$\bar{Y}$
$P(Y\|\bar{D})\cdot 992 =?$	$P(Y\|D)\cdot 8 =?$	Y

and

$\bar{D}$	D	
$P(\bar{D}\|\bar{Y})\cdot 923 =?$	$P(D\|\bar{Y})\cdot 923 =?$	$\bar{Y}$
$P(\bar{D}\|Y)\cdot 77 =?$	$P(D\|Y)\cdot 77 =?$	Y

Bayes Theorem Example

We can use the theorem to find the probability our patient has the disease, given the positive test result:

$$\begin{aligned} P(D|Y) &= P(Y|D)\,\frac{P(D)}{P(Y)} \\ &= 0.90 \cdot \frac{0.008}{0.077} \\ &= 0.094. \end{aligned}$$

That is, there's only a one-in-eleven chance the patient actually has the disease.

Bayes Theorem Example

We have two conditional probabilities given: $P(Y|D)$ and $P(Y|\bar{D})$. We found $P(D|Y)$. What about $P(D|\bar{Y})$? We can use the partition theorem (theorem **??**) again to solve for what we don't know in terms of what we do know:

$$\begin{aligned} P(D) &= P(D|Y)\,P(Y) + P(D|\bar{Y})\,P(\bar{Y}) \\ P(D|\bar{Y}) &= \frac{P(D) - P(D|Y)\,P(Y)}{P(\bar{Y})} \\ &= \frac{0.008 - 0.094 \cdot 0.077}{0.923} \\ &= 0.0008. \end{aligned}$$

We now have all four conditional probabilities:

$$\begin{aligned} P(Y|\bar{D}) &= 0.07 & \qquad P(D|\bar{Y}) &= 0.0008 \\ P(Y|D) &= 0.90 & \qquad P(D|Y) &= 0.094. \end{aligned}$$

Bayes Theorem Example

Now we can fill out the four-square table:

- Since $P(Y|\bar{D}) = 0.07$, seven percent of the 992 disease-free people (70 of them) get false positives; the rest (922) get a correct negative result.
- Since $P(Y|D) = 0.90$, ninety percent of the 8 people with the disease test positive (i.e. all but one of them); one of the 8 gets a false sense of security.
- Since $P(D|\bar{Y}) = 0.0008$, 0.08% of the 923 with negative test results (one person) does in fact have the disease; the other 922 (as we found just above) get a correct negative result.
- Since $P(D|Y) = 0.094$, only 9.4% of the 77 people with positive test results (7 people) have the disease; the other 70 get a scare (and, presumably, a re-test).

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Bayes Theorem Example

So, the sample of 1000 people splits up as follows:

$\bar{D}$	D	
922	1	$\bar{Y}$
70	7	Y

Moreover, we can rank events by likelihood:

(1) Healthy people correctly diagnosed: 92.2%.

(2) False positives: 7%.

(3) People with the disease, correctly diagnosed: 0.7%.

(4) False negatives: 0.1%.

Now it's no surprise our patient got a false positive: this happens 10 times as often as a correct positive diagnosis.

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Comparisions

- Classical statistics: model parameters are *fixed* and *unknown*.
- A Bayesian thinks of parameters as random, and thus having distributions (just like the data). We can thus think about unknowns for which no reliable frequentist experiment exists, e.g. $\theta =$ proportion of US men with untreated prostate cancer.
- A Bayesian writes down a *prior* guess for parameter(s) θ, say $p(\theta)$. He then combines this with the information provided by the observed data *by* to obtain the *posterior* distribution of θ, which we denote by $p(\theta givenby)$.
- All statistical inferences (point and interval estimates, hypothesis tests) then follow from posterior summaries. For example, the posterior means/medians/modes offer point estimates of θ, while the quantiles yield credible intervals.

- The key to Bayesian inference is "learning" or "updating" of prior beliefs. Thus, posterior information $\geq$ prior information.

- Is the classical approach wrong? That may be a controversial statement, but it certainly is fair to say that the classical approach is limited in scope.

- The Bayesian approach expands the class of models and easily handles:
  - repeated measures
  - unbalanced or missing data
  - nonhomogenous variances
  - multivariate data

  – and many other settings that are precluded (or much more complicated) in classical settings.

YHK

Thanks ...

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